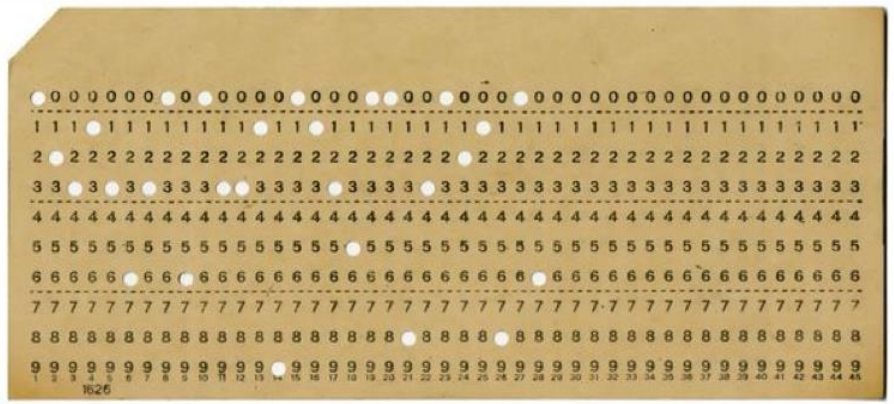
EVOLUTION OF OS

1. Batch Operating System
2. Multiprogramming Operating System
3. Time-Sharing OS
4. Multiprocessing OS
5. Distributed OS
6. Network OS
7. Real Time OS
8. Embedded OS

Serial Processing

- **No operating system**
  - Programmers interacted directly with the computer hardware
- **Computers ran from a console with display lights, toggle switches, some form of input device, and a printer**
- **Users have access to the computer in "series"**



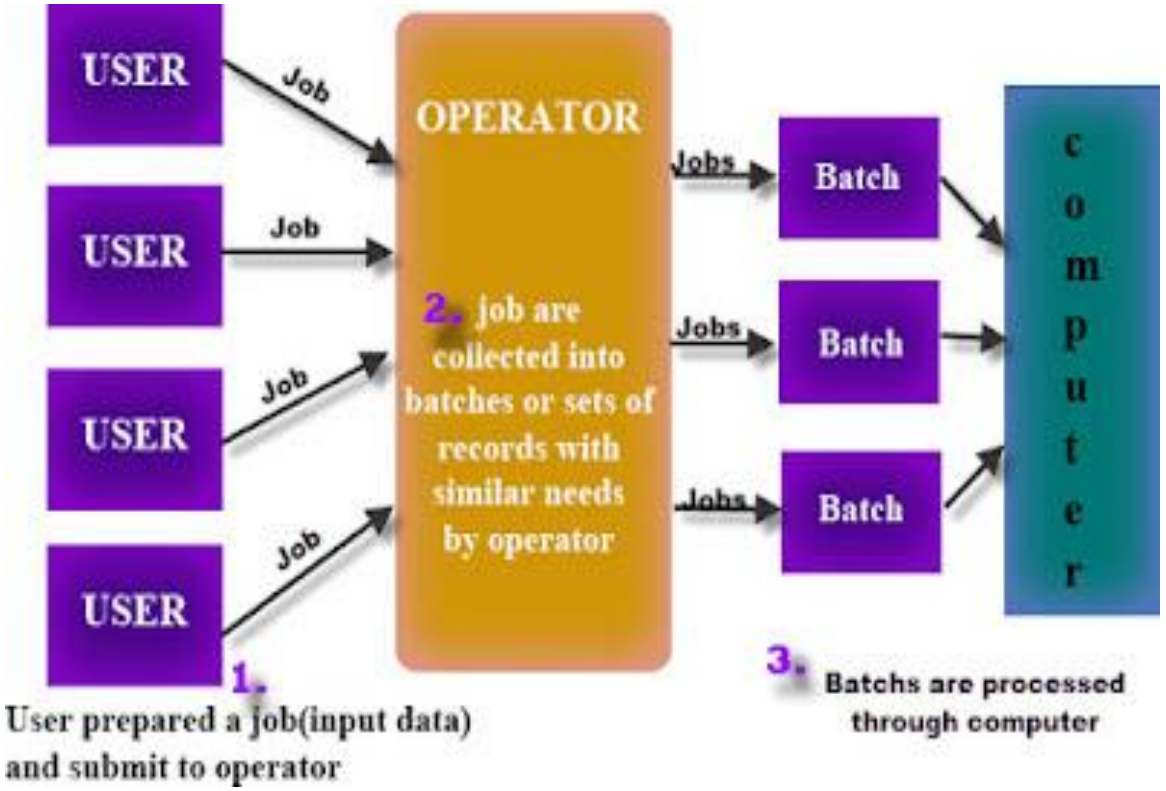
Problems:

- **Scheduling :**
  - Most installations used a hardcopy sign-up sheet to reserve computer time
  - Time allocations could run short or long, resulting in wasted computer time
- **Setup time :**
  - A considerable amount of time was spent on setting up the program to run

- *Early computers were very expensive*
- *Important to maximize processor utilization*

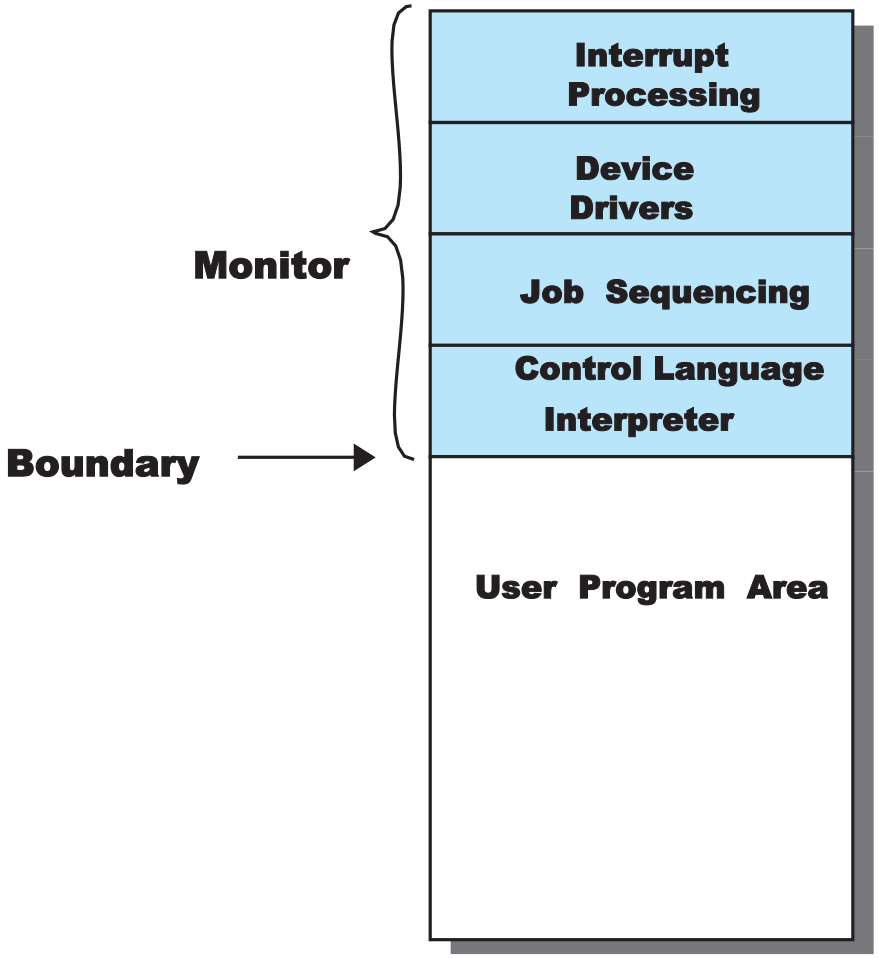
Simple Batch Systems

- Monitor
  - User no longer has direct access to processor
  - Job is submitted to computer operator who batches them together and places them on an input device
  - Program branches back to the monitor when finished



Monitor Point of View

- Monitor controls the sequence of events
- Resident Monitor is software always in memory
- Monitor reads in job and gives control
- Job returns control to monitor



Memory Layout for a Resident Monitor

Processor Point of View

- Processor executes instruction from the memory containing the monitor
    - Executes the instructions in the user program until it encounters an ending or error condition
- ***"Control is passed to a job"*** means processor is fetching and executing instructions in a user program
- ***"Control is returned to the monitor"*** means that the processor is fetching and executing instructions from the monitor program

Job Control Language (JCL)

Special type of programming language used to provide instructions to the monitor

What compiler to use

What data to use

Desirable Hardware Features

Memory protection

- **While the user program is executing, it must not alter the memory area containing the monitor**

Timer

- **Prevents a job from monopolizing the system**

Privileged instruction

- **Can only be executed by the monitor**

Interrupts

- **Gives OS more flexibility in controlling user programs**

Simple Batch System Overhead

- Processor time alternates between execution of user programs and execution of the monitor
- Sacrifices:
  - Some main memory is now given over to the monitor
  - Some processor time is consumed by the monitor
- Despite overhead, the simple batch system improves utilization of the computer

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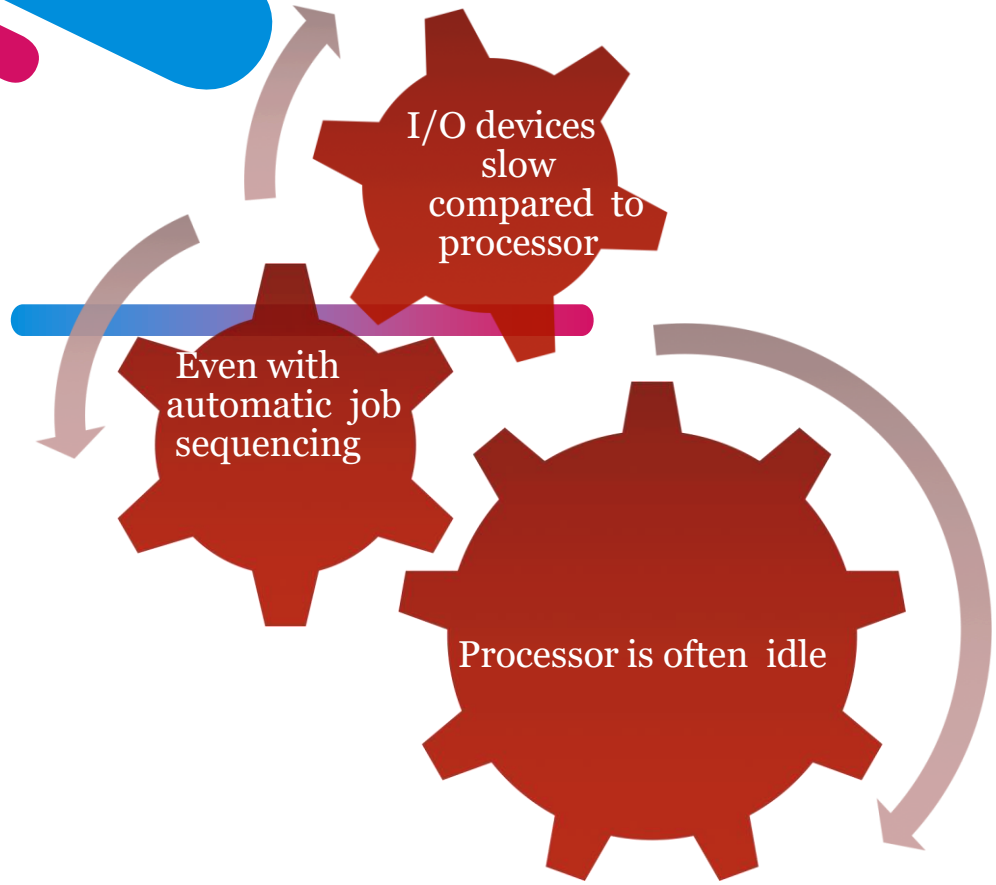
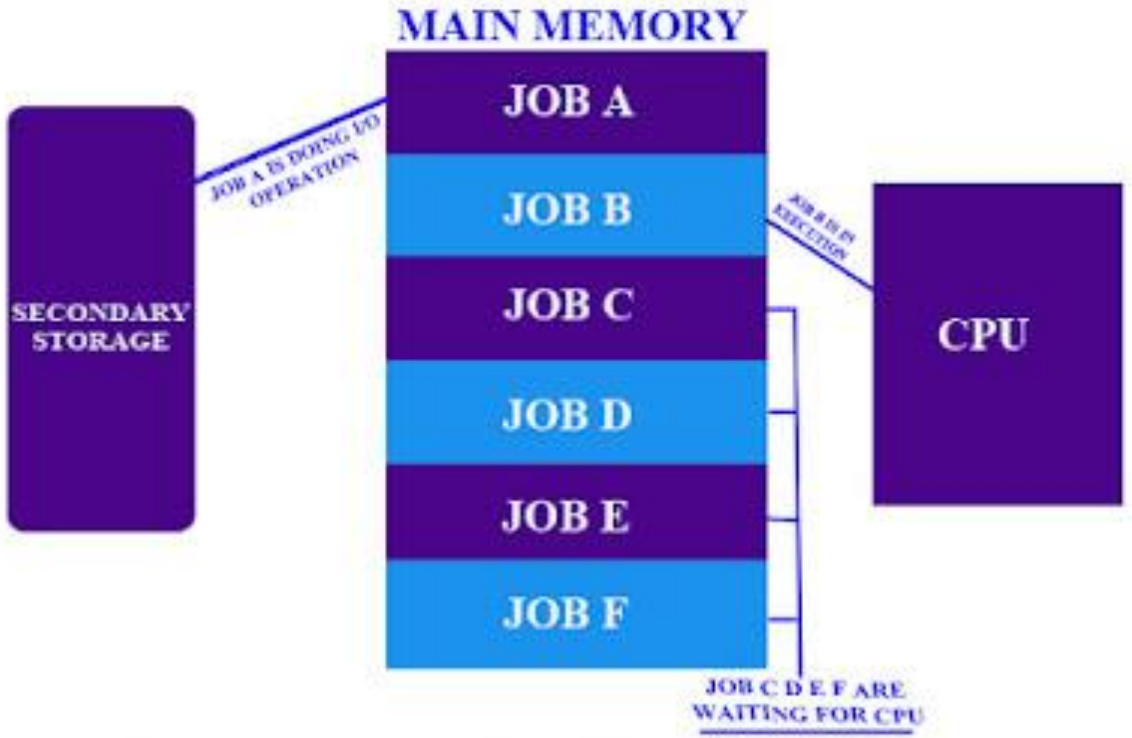
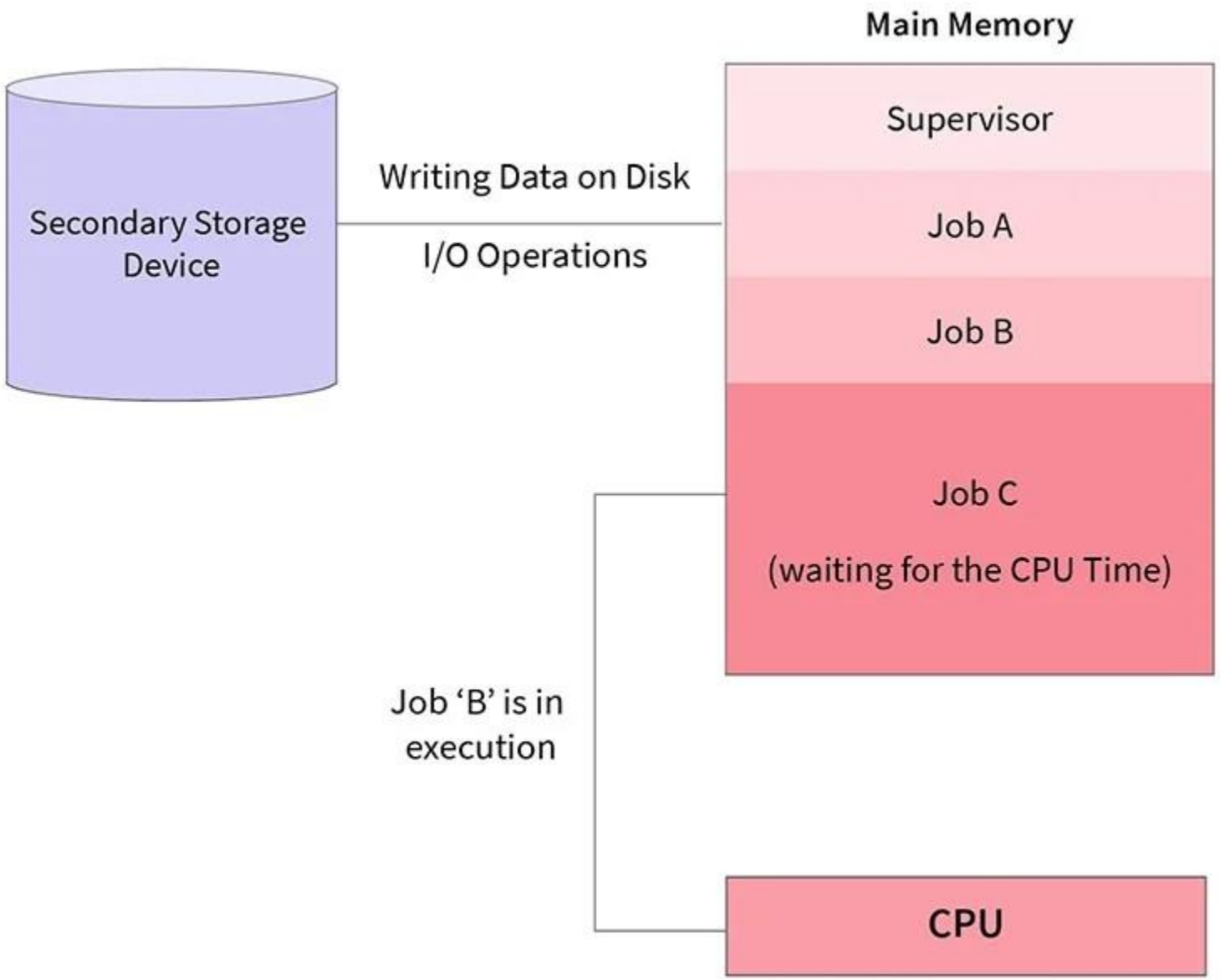
Multiprogramming Systems





Multiprogramming Operating System


Main Memory
Supervisor
Job A
Job B
Job C
(waiting for the CPU Time)
Secondary Storage Device
Writing Data on Disk
I/O Operations
Job 'B' is in execution
CPU

Multitasking OS: Enables execution of multiple programs at the same time. The operating system accomplishes this by swapping each program in and out of memory one at a time. When a program is switched out of memory, it is temporarily saved on disk until it is required again.

Multiuser Operating System: This allows many users to share processing time on a powerful central computer from different terminals. The operating system accomplishes this by rapidly switching between terminals, each of which receives a limited amount of processor time on the central computer

Advantages:

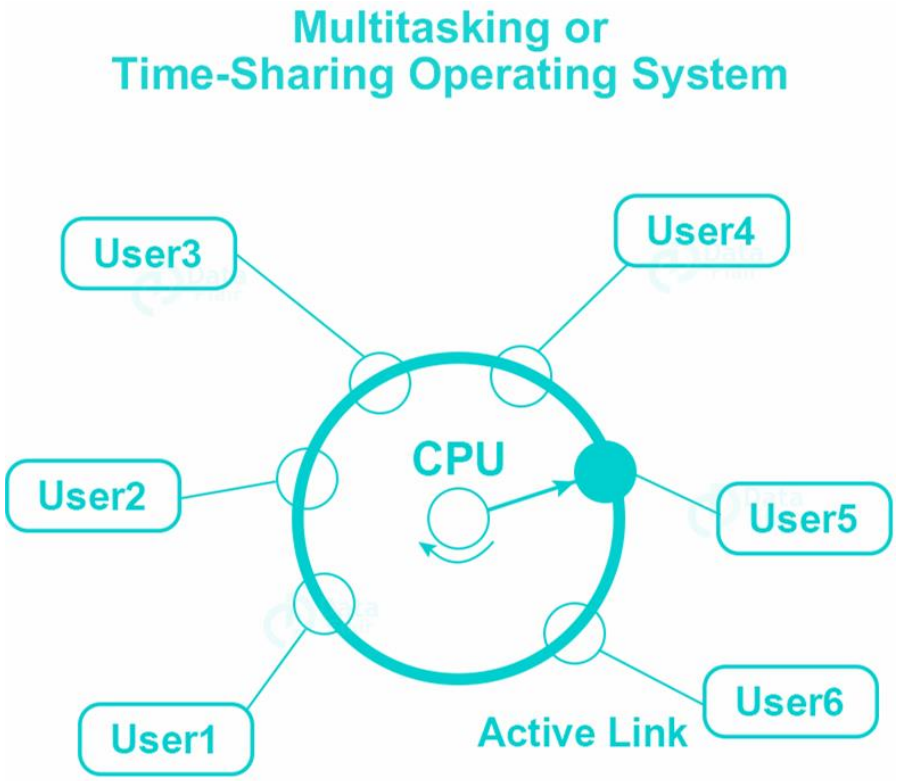
- It provides high and efficient CPU utilization.
- It gives an illusion to the users that multiple programs are allotted CPU almost simultaneously.

Disadvantages:

- CPU scheduling is necessary to perform multiprogramming.
- Memory management accommodates multiple jobs in the memory.

Time Sharing Systems

- Can be used to handle multiple interactive jobs
- The computer memory stores these programs and they share a single processor.
- The tasks switch rapidly from one to another giving immediate response to the user(s).
- It uses the concept of CPU scheduling and multiprogramming so that the time of the CPU is divided equally among the tasks and decreasing the response time of a task.
- The time taken to execute a single task is called <u>quantum</u>. Once this time interval is over the OS switches to the next task.



Advantages of Time-Sharing System

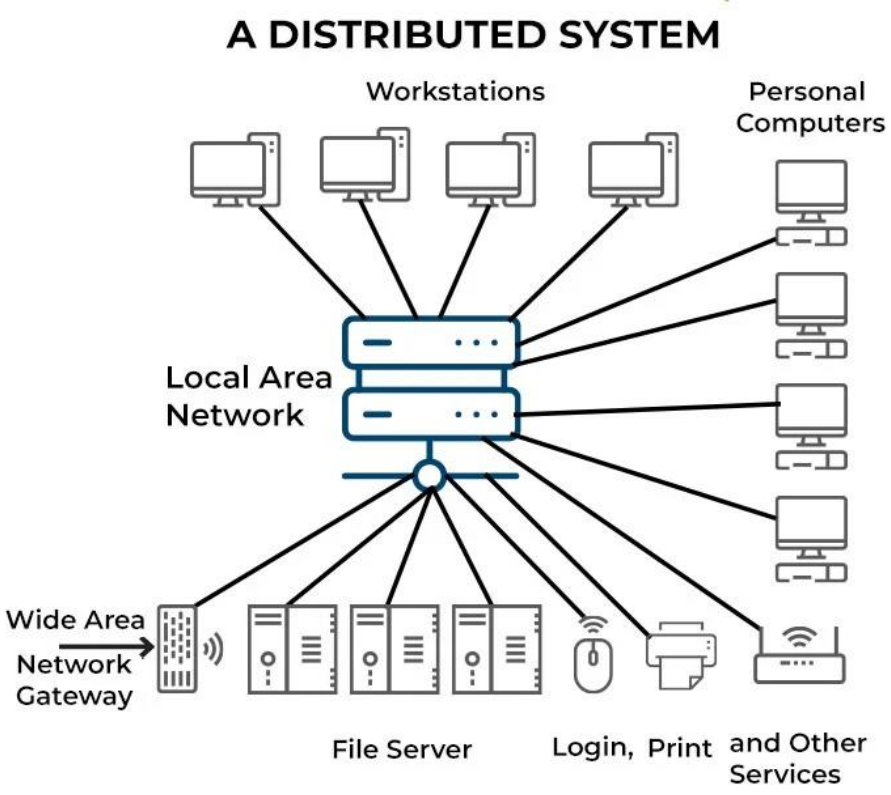
- The Time-sharing system offers the benefit of quick response.
- The Time-sharing system minimizes the idle time of the CPU.
- The Time-sharing system avoids the duplication of the software.
- The Time-sharing system provides efficient utilization of CPU.

Disadvantages of Time-Sharing System

- In the Time-sharing system, there is a problem of reliability.
- Data communication problem

Distributed operating system

- A distributed operating system allows the distribution of entire systems on the couples of center processors, and it serves on multiple real-time products as well as multiple users
- The Distributed operating systems are also known as loosely coupled systems.



Advantages

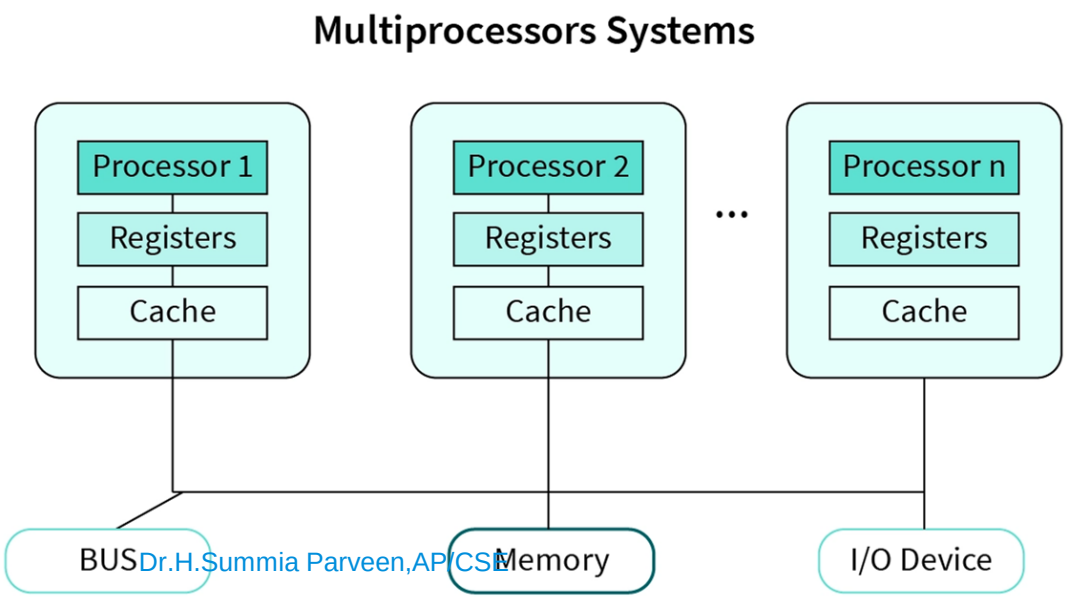
- A distributed operating system may share all resources from one site to another, increasing data availability across the entire system.
- It reduces the probability of data corruption because all data is replicated across all sites.
- The entire system operates independently of one another, and as a result, if one site crashes, the entire system does not halt.
- A distributed operating system is an open system since it may be accessed from both local and remote locations.
- It helps in the reduction of data processing time.

Disadvantages

- The system must decide which jobs must be executed when they must be executed, and where they must be executed. A scheduler has limitations, which can lead to underutilized hardware and unpredictable runtimes.
- It is hard to implement adequate security in a distributed operating system since the nodes and connections must be secured.
- The database connected to a DOS is relatively complicated and hard to manage in contrast to a single-user system.
- The underlying software is extremely complex and is not understood very well compared to other systems.
- The more widely distributed a system is, the more communication latency can be expected. As a result, teams and developers must choose between availability, consistency, and latency.
- Gathering, processing, presenting, and monitoring hardware use metrics for big clusters can be a real issue.

Multiprocessor Operating System

- Multiprocessor operating system utilizes multiple processors, which are connected with physical memory, computer buses, clocks, and peripheral devices (touchpad, joystick, etc). The main objective of using a multiprocessor OS is to consume high computing power and increase the execution speed of the system.



CPU – capable to access memories as well as controlling the entire I/O tasks.

Input Output Processor – I/P processor can access direct memories, and every I/O processors have to be responsible for controlling all input and output tasks.

Input/Output Devices – These devices are used for inserting the input commands, and producing output after processing.

Memory Unit – Multiprocessor system uses the two types of memory modules - shared memory and distributed shared memory.

Advantages

- Great Reliability.
- Improve Throughput.
- Cost-Effective System.
- Parallel Processing.

Disadvantages

It is more expensive due to its large architecture.

Its speed can get degraded due to failing any one processor.

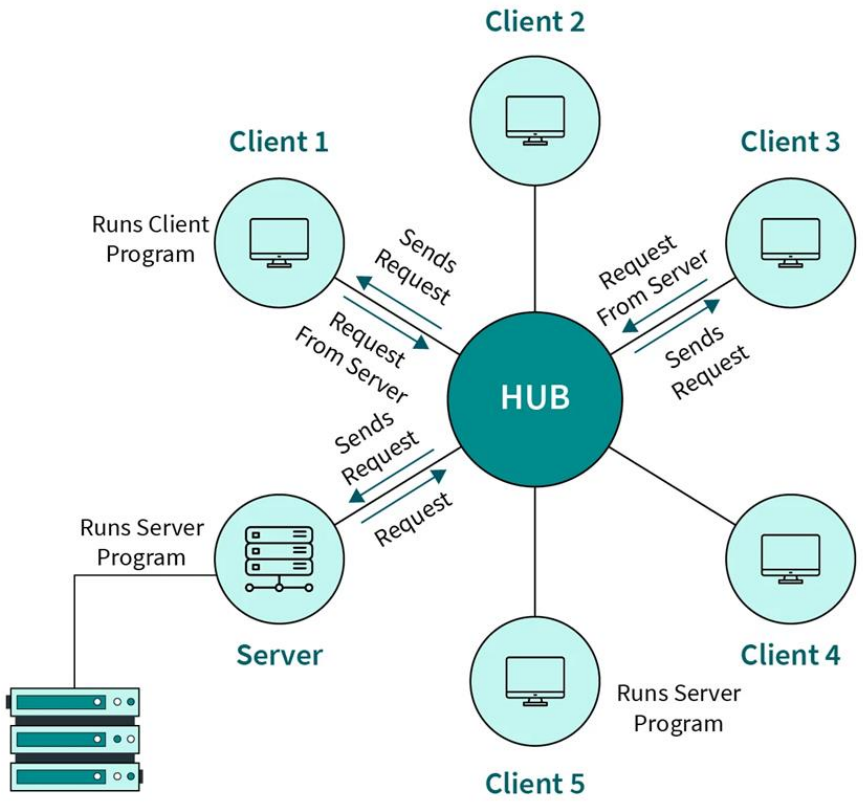
It has more time delay when the processor receives the message and takes appropriate action.

It has big challenges related to skew and determinism.

It needs context switching which can be impacted its performance.

Network Operating System

- Network Operating System has special functions for connecting computers and devices into a local-area network or Inter-network. Some popular network operating systems are Novell Netware, Linux, IBM OS/2, etc.
- The network operating system which was first developed is Novell Netware.



Client Server Network

- There are two basic types of network operating systems, discussed as follows:

Peer-to-Peer Network Operating Systems: allow users to share network resources saved in a common, accessible network location.

Client/Server Network Operating Systems: provide users with access to resources through a server.

Advantages

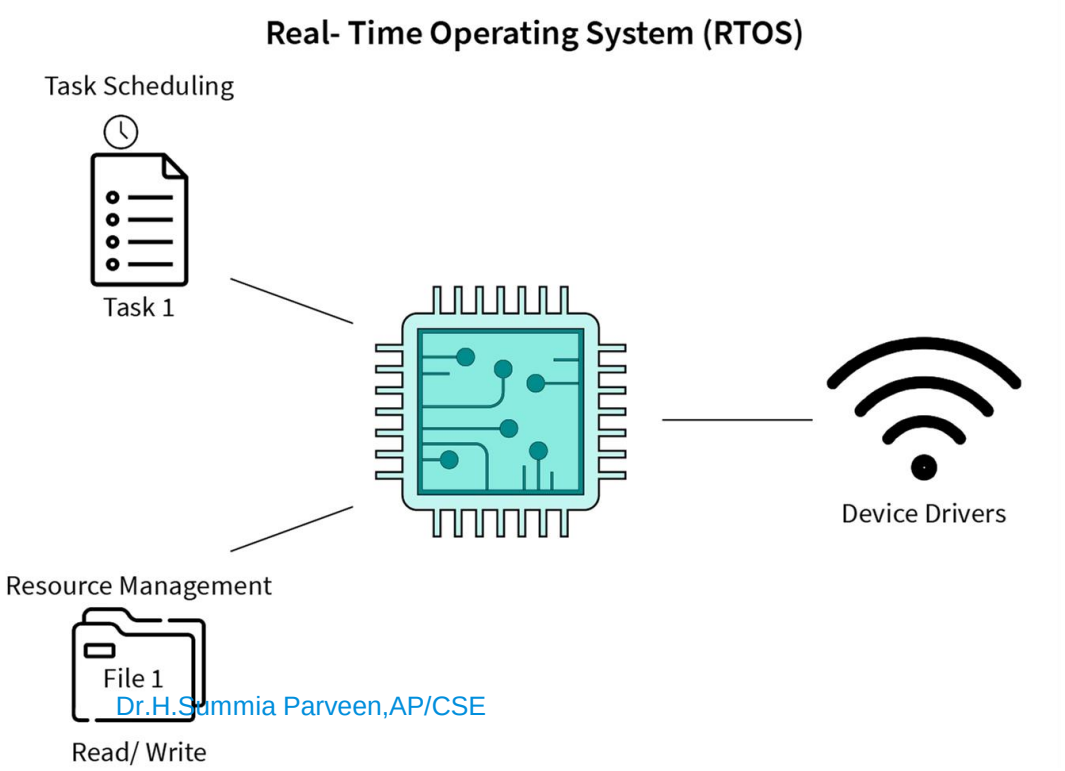
- Highly stable centralized servers
- Security concerns are handled through servers
- New technologies and hardware up-gradation are easily integrated into the system
- Server access is possible remotely from different locations and types of systems

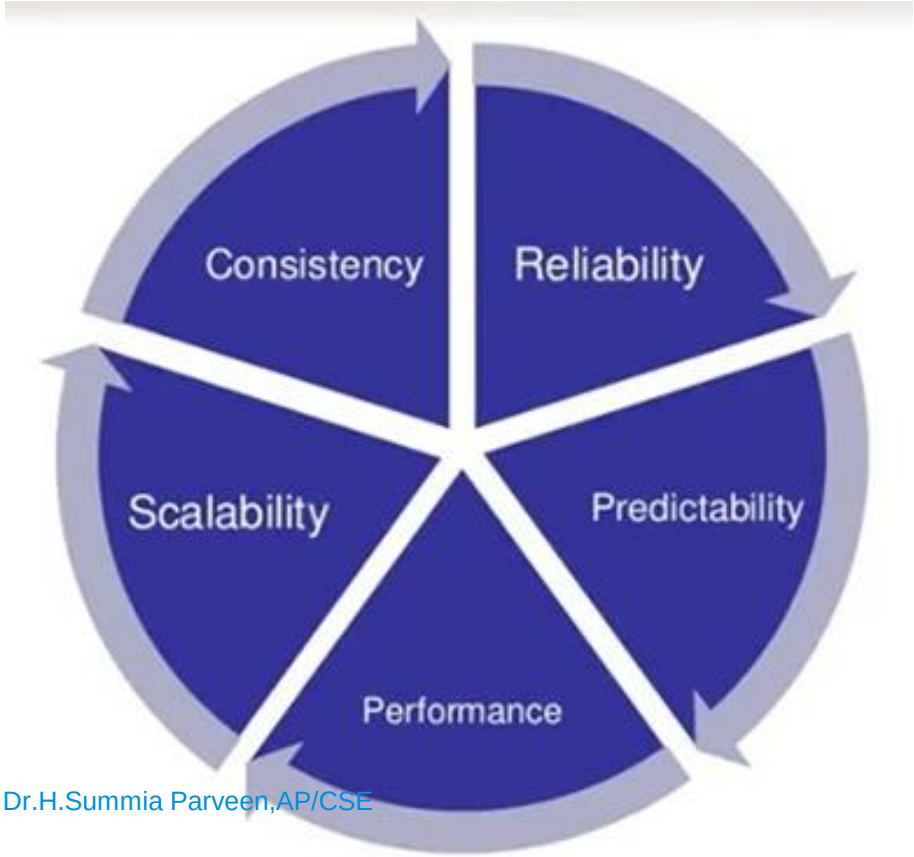
Disadvantages

- Servers are costly
- User has to depend on a central location for most operations
- Maintenance and updates are required regularly

Real Time Operating System

A real time system is used when there is a rigid time requirement on the operation of a processor or the flow of data and thus is often used as a control device in a delicated application system. The system that controls scientific experiments, industrial control systems, and some display systems are real time systems.




Consistency
Reliability
Scalability
Predictability
Performance

Advantages

Priority-Based Scheduling.

Abstracting Timing Information.

Maintainability/Extensibility.

Modularity.

Promotes Team Development.

Easier Testing.

Code Reuse.

Improved Efficiency.

Idle Processing.

Disadvantages

Limited Tasks.
Use Heavy System resources.
Complex Algorithms.
Device driver and interrupt signals.
Thread Priority.